Christian Andersson Naesseth

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Employment

Assistant Professor (UD1, Tenured) — University of Amsterdam
Amsterdam Machine Learning Lab — Jan 2025 – Present

Assistant Professor (UD2, Tenure-Track) — University of Amsterdam
Amsterdam Machine Learning Lab — Jan 2022 – Dec 2024

Postdoctoral Research Scientist — Columbia University
Data Science Institute — Aug 2019 – Dec 2021
Advisor: David M. Blei

Postdoctoral Researcher — Linköping University
Department of Computer and Information Science — Jan 2019 – Jul 2019
Advisor: Fredrik Lindsten

Research Intern — Microsoft Research Ltd
Machine Intelligence & Perception — Apr 2018 – Jul 2018
Supervisor: Sebastian Nowozin

Fulbright Visiting Student Researcher — Columbia University
Data Science Institute — Jun 2016 – Jul 2017
Advisor: David M. Blei

Education

Ph.D. Electrical Engineering with Specialization in Automatic Control — 2019
Linköping University, Sweden
Dissertation: *Machine learning using approximate inference: Variational and SMC methods*
Advisors: Thomas B. Schön, Fredrik Lindsten

M.Sc. Applied Physics and Electrical Engineering — 2013
Linköping University, Sweden
Thesis: *Vision and Radar Sensor Fusion for Advanced Driver Assistance Systems*

B.Sc. Mathematics — 2012
Linköping University, Sweden
Thesis: *Nowcasting using Microblog Data*

Honors and Awards

Best Paper Award — 2025
Symposium on Advances in Approximate Bayesian Inference (AABI)
SDE Matching: Scalable and Simulation-Free Training of Latent Stochastic Differential Equations

Savage Award (Theory and Methods) — 2019
International Society for Bayesian Analysis (ISBA)
Best PhD dissertation in Bayesian statistics
Machine learning using approximate inference: Variational and sequential Monte Carlo methods

Best Paper Award — 2017
International Conference on Artificial Intelligence and Statistics (AISTATS)
Reparameterization Gradients through Acceptance–Rejection Algorithms

Fulbright Scholarship 2016
Fulbright Commission

Research Scholarships 2016
Ericsson Research Foundation, Gålöstiftelsen, Bernt Järmarks stiftelse

Best Poster Award 2014
Summer School on Deep Learning for Image Analysis
Sequential Monte Carlo for Graphical Models

Research Funding and Program Leadership

Small Compute 2026 – 2027
- Fork-dLLM: diversity-enhanced parallel sampling in masked diffusion language models
- Learning Sampling in Diffusion LLMs

Funding source: NWO
Role: PI

Generative models and uncertainty quantification in machine learning 2025 – 2026
Gift funding for 1 postdoctoral researcher.
Funding source: Bosch & Scyfer
Role: PI

UvA-Bosch Delta Lab 2 2023 – 2027
Industry-funded research program with Bosch Group supporting 10 PhD students.
Funding source: Bosch & RVO
Role: Co-I, Lab Manager

Academic Supervision

PhD Candidates

Rajeev Verma (with Eric Nalisnick, Volker Fischer) 2023 –
University of Amsterdam

Alexander Timans (with Eric Nalisnick, Kaspar Sakmann, Christoph-Nikolas Straehle) 2022 –
University of Amsterdam

Metod Jazbec (with Eric Nalisnick, Dan Zhang, Stephan Mandt) 2022 –
University of Amsterdam

Grigory Bartosh 2022 –
University of Amsterdam

Mona Schirmer (with Eric Nalisnick, Dan Zhang) 2022 –
University of Amsterdam

Heiko Zimmermann (with Jan-Willem van de Meent) 2021 – 2025
University of Amsterdam
Simulation Intelligence Scientist, Pasteur Labs & ISI

Postdocs

Hany Abdulsamad 2025 –
University of Amsterdam

Visiting PhD Students

University of Amsterdam: Ziwei Luo (2026), Christopher Marouani (2026), Kenny Falkær Olsen (2026), Alejandro Valverde Mahou (2026), Markus Müller (2025-2026), Fabian Denoodt (2025), Bahrul Nasution (2025), Raghuram D R (2024), François Cornet (2024).

Master Students

University of Amsterdam: Maria Marchenko (2026), Odysseas Boufalis (2026), Rohith Prabakaran (2025), Aditya Patra (2025), Nesta Midavaine (2025), Doris Wezenberg (2024).

Linköping University: Elina Fantenberg (2018), Martin Lindfors (2014), Olle Noren (2014), Alfred Dahlin (2014).

Invited Talks

Simulation-Free Variational Inference for SDEs 2026
Department of Mathematical Sciences (Seminar) Chalmers

Generative Modelling for Dynamical Systems 2026
Workshop on Scientific Machine Learning and Uncertainty Quantification AIMET-NL/CWI

Scalable Variational Inference for SDEs 2025
Workshop on industrial applications of numerical analysis and machine learning CWI

Generative Modeling and Inverse Problems in Molecule Generation 2025
ChemAI: AI-powered chemical and material innovations ChemAI

SDE Matching: Scalable Variational Inference for SDEs 2025
The Royal Swedish Academy Of Sciences, AI4Science Symposium KVA

SDE Matching 2025
BIRS Workshop: Efficient Approximate Bayesian Inference BIRS

Neural Flow Diffusion Models and SDE Matching 2025
MFO Mini-Workshop on Statistical Challenges for Deep Generative Models MFO

Diffusions, flows, and other stories 2024
NeurIPS Fest (Keynote) University of Amsterdam

There And Back Again: A Forward Diffusion Tale 2024
Generative models and uncertainty quantification GenU

Generative Models and Approximate Bayesian Inference 2024
Special Invited Session: Bayesian computational methods COMPSTAT

There And Back Again: A Diffusion's Tale 2024
Industry-on-Campus Lab (Seminar) Bosch Center for AI and University of Tübingen

Twisted Diffusion Sampling for Accurate Conditional Generation 2023
Plenary talk ELLIS unConference

Monte Carlo and Variational Methods: Bridging the Gap 2022
Special Invited Session: Grand challenges and advances in Bayesian computation CMStatistics
Workshop on Monte Carlo and Approximate Dynamic Programming Methods ESSEC Paris

Variational Bayes Goes to Monte Carlo 2021
Amsterdam Machine Learning lab (Seminar) University of Amsterdam

Machine learning using approximate inference 2020
Savage Award session (Contributed Talk) Joint Statistical Meeting
Junior Bayes Beyond the Borders (Webinar) Bocconi University

Variational and Monte Carlo methods 2019
Center for Industrial and Applied Mathematics (Seminar) KTH
Department of Mathematical Sciences (Seminar) Chalmers

Variational inference 2018
Department of Information Technology (Tutorial) Uppsala University

Approximate Bayesian inference: Variational and MC methods 2017
Department of Computer Science (Seminar) Linköping University

Monte Carlo methods and proper weighting 2015
Department of Engineering Science (Tutorial) The University of Oxford

Nested Sequential Monte Carlo Methods 2015
SMC Workshop ENSAE Paris

Sequential Monte Carlo for Probabilistic Graphical Models 2014
School of Mathematics and Statistics (Seminar) University of NSW
School of Electrical Engineering and Computer Science (Seminar) University of Newcastle

Teaching

Machine Learning (Undergraduate) 2025 –
Lecturer University of Amsterdam

Reinforcement Learning (Graduate) 2024 – 2026
Lecturer University of Amsterdam

Introduction to Machine Learning (Undergraduate) 2022 – 2024
Lecturer University of Amsterdam

Digital Expertise: Introduction to ML (Undergraduate) 2024
Guest lecturer University of Amsterdam

Foundations of Graphical Models (Graduate) 2019
Guest lecturer Columbia University

Teaching Assistant and Project Supervisor 2011 – 2018
Linköping University

Contributed to undergraduate and graduate courses in engineering, control, signal processing, sensor fusion, and mathematics. Responsibilities included recitation instruction, laboratory teaching, project supervision, and grading.

Professional Service

Organisation

Symposium on Probabilistic Machine Learning 2026
Co-organizer, Advisor ProbML

Symposium on Advances in Approximate Bayesian Inference 2023 – 2025
Co-organizer, Program Chair, Sponsorship Chair AABI

International Conference on Artificial Intelligence and Statistics 2023
Workflow Chair AISTATS

Senior Program Committee

International Conference on Artificial Intelligence and Statistics 2024 – 2026
Senior Area Chair AISTATS

International Conference on Machine Learning 2027
Area Chair ICLR

International Conference on Machine Learning 2026
Area Chair ICML

Conference on Neural Information Processing Systems 2025 – 2026
Area Chair NeurIPS

Conference on Uncertainty in Artificial Intelligence 2024, 2026
Area Chair UAI

International Conference on Artificial Intelligence and Statistics 2022
Area Chair AISTATS

Reviewing

Dutch Research Council (NWO) 2024

Journal of Machine Learning Research (JMLR) 2020 – 2021

Neural Information Processing Systems (NeurIPS, EurIPS) 2017 – 2020, 2025
Best Reviewer Award NeurIPS 2017

International Conference on Machine Learning (ICML) 2017 – 2018

International Conference on Learning Representations (ICLR) 2017

International Conference on Artificial Intelligence and Statistics (AISTATS) 2017 – 2018

Doctoral Committees

Angus Phillips 2026
Deep Learning Methods for Intractable Distributions in Sampling, Generative Modelling and Experimental Design — Oxford University

Maxence Noble 2026
Sampling and Transport with Score-based Diffusion Models: Opportunities and Methodological Advances — École Polytechnique

Kasper Bågmark 2026
Learning to Estimate: Bayesian Filtering with Deep Density Methods — Chalmers University of Technology

Fiona Lippert 2025
From weather radars to bird migration fluxes: Process-guided machine learning for spatio-temporal forecasting and inference — University of Amsterdam

Gabriel Bénédict 2024
A Machine Learning Personalization Flow — University of Amsterdam

Salem Lahlou 2023
Advances in uncertainty modelling: from epistemic uncertainty estimation to generalized generative flow networks — Université de Montréal, MILA

Professional Development

Superb Supervision 2025
Mennen Training & Consultancy — University of Amsterdam

University Teaching Qualification (BKO) 2024
University of Amsterdam

Leadership Course for Tenure Trackers 2022
Center for Academic Leadership — University of Amsterdam

Learning and Knowledge 2016
Advanced course in higher education pedagogy — Linköping University

Publications

S. Frkovic*, M. Jazbec*, D. Zhang, C. A. Naesseth, I. Bogunovic, and E. Nalisnick. Re-evaluating confidence remasking in masked diffusion language models. *arXiv:2606.12232*, 2026.

H. Abdulsamad, S. Iqbal, C. A. Naesseth, T. Matsubara, and A. Corenflos. Maximin robust Bayesian experimental design. *arXiv:2603.14094*, 2026.

T. X. Olausson, M. Jazbec, X. Wang, A. Solar-Lezama, C. A. Naesseth, S. Mandt, and E. Nalisnick. A tale of two temperatures: Simple, efficient, and diverse sampling from diffusion language models. In *Third Conference on Language Modelling (COLM)*, 2026.

A. Timans, T. Möllenhoff, C. A. Naesseth, M. Emtiyaz Khan, and E. Nalisnick. Joint model and data sparsification via the marginal likelihood. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.

M. Schirmer, M. Jazbec, A. Timans, C. A. Naesseth, M. Waldron, and E. Nalisnick. Online safety monitoring for LLMs. In *ICML 2026 Workshop on Hypothesis Testing*, 2026.

R. Saai Pemmasani Prabakaran, S. Verdenius, P. Das, and C. A. Naesseth. On the role of context in zero-shot multivariate time-series forecasting. In *ICLR 2026 Workshop on Time Series in the Age of Large Models (TSALM)*, 2026.

L. Y. Chanchu, H. Abdulsamad, and C. A. Naesseth. Discrete diffusion inference-time control with nested sequential Monte Carlo. In *ICLR 2026 Workshop on on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models (ReALM-GEN)*, 2026.

G. Kerrigan, C. A. Naesseth, and T. Rainforth. A geometric approach to optimal experimental design. In *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.

B. I. Nasution, F. Eijkelboom, M. Elliot, R. Allmendinger, and C. A. Naesseth. Flow matching for tabular data synthesis. *Transactions on Machine Learning Research*, 2026.

N. Midavaine, C. A. Naesseth, and G. Bartosh. Towards latent diffusion suitable for text. In *EurIPS 2025 Workshop on Principles of Generative Modeling (PriGM)*, 2025.

R. Verma, R. Derr, C. A. Naesseth, V. Fischer, and E. Nalisnick. So what are good imprecise forecasts? In *EurIPS 2025 Workshop on Epistemic Intelligence in Machine Learning (EIML)*, 2025.

L. Wu, Y. Han, C. A. Naesseth, and J. P. Cunningham. Reverse diffusion sequential Monte Carlo samplers. In *Advances in Neural Information Processing Systems (NeurIPS) 38*, 2025.

M. Schirmer*, M. Jazbec*, C. A. Naesseth, and E. Nalisnick. Monitoring risks in test-time adaptation. In *Advances in Neural Information Processing Systems (NeurIPS) 38*, 2025. * equal contribution.

A. Timans*, R. Verma*, E. Nalisnick, and C. A. Naesseth. On continuous monitoring of risk violations under unknown shift. In *Uncertainty in Artificial Intelligence (UAI)*, 2025. * equal contribution.

G. Bartosh, D. Vetrov, and C. A. Naesseth. SDE Matching: Scalable and simulation-free training of latent stochastic differential equations. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Vancouver, Canada, Jul 2025. **(Best Workshop Paper Award at AABI 2025)**.

F. Eijkelboom, H. Zimmermann, S. Vadgama, E. J. Bekkers, M. Welling, J-W. van de Meent*, and C. A. Naesseth*. Controlled generation with equivariant variational flow matching. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Vancouver, Canada, Jul 2025. * equal contribution.

A. Timans, C.-N. Straehle, K. Sakmann, C. A. Naesseth, and E. Nalisnick. Max-rank: Efficient multiple testing for conformal prediction. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.

A. Chen, P. Chlenski, K. Munyuza, A. K. Moretti, C. A. Naesseth, and I. Pe'er. Variational combinatorial sequential Monte Carlo for Bayesian phylogenetics in hyperbolic space. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.

F. Cornet, G. Bartosh, M. Schmidt, and C. A. Naesseth. Equivariant neural diffusion for molecule generation. In *Advances in Neural Information Processing Systems (NeurIPS) 37*, 2024.

F. Eijkelboom*, G. Bartosh*, C. A. Naesseth, M. Welling, and J-W. van de Meent. Variational flow matching for graph generation. In *Advances in Neural Information Processing Systems (NeurIPS) 37*, 2024. * equal contribution.

H. Yang, A. K. Moretti, S. Macaluso, P. Chlenski, C. A. Naesseth, and I. Pe'er. Variational pseudo marginal methods for jet reconstruction in particle physics. *Transactions on Machine Learning Research*, 2024.

M. Jazbec*, A. Timans*, T. H. Veljković, K. Sakmann, D. Zhang, C. A. Naesseth, and E. Nalisnick. Fast yet safe: Early-exiting with risk control. In *Advances in Neural Information Processing Systems (NeurIPS) 37*, 2024. * equal contribution.

G. Bartosh, D. Vetrov, and C. A. Naesseth. Neural flow diffusion models: Learnable forward process for improved diffusion modelling. In *Advances in Neural Information Processing Systems (NeurIPS) 37*, 2024a.

H. Zimmermann, C. A. Naesseth, and J-W. van de Meent. VISA: Variational inference with sequential sample-average approximations. In *Advances in Neural Information Processing Systems (NeurIPS) 37*, 2024.

G. Bartosh, D. Vetrov, and C. A. Naesseth. Neural diffusion models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, Vienna, Austria, Jul 2024b.

T. Pandeva, T. Bakker, C. A. Naesseth, and P. Forré. E-valuating classifier two-sample tests. *Transactions on Machine Learning Research*, 2024.

L. Wu, B. L. Trippe, C. A. Naesseth, D. M. Blei, and J. P. Cunningham. Practical and asymptotically exact conditional sampling in diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS) 36*, 2023.

L. Zhang, D. Blei, and C. A. Naesseth. Transport score climbing: Variational inference using forward KL and adaptive neural transport. *Transactions on Machine Learning Research*, 2023.

H. Zimmermann, F. Lindsten, J-W. van de Meent, and C. A. Naesseth. A variational perspective on generative flow networks. *Transactions on Machine Learning Research*, 2023.

A. K. Moretti, L. Zhang, C. A. Naesseth, H. Venner, D. Blei, and I. Pe'er. Variational combinatorial sequential Monte Carlo methods for Bayesian phylogenetic inference. In *Uncertainty in Artificial Intelligence (UAI)*, 2021.

C. A. Naesseth, F. Lindsten, and D. Blei. Markovian score climbing: Variational inference with KL(p||q). In *Advances in Neural Information Processing Systems (NeurIPS) 33*, Vancouver, Canada, 2020.

D. Biderman, C. A. Naesseth, L. Wu, T. Abe, A. C. Mosberger, L. J. Sibener, R. M. Costa, J. Murray, and J. Cunningham. Inverse articulated-body dynamics from video via variational sequential Monte Carlo. In *First workshop on differentiable computer vision, graphics, and physics in machine learning (NeurIPS)*, Vancouver, Canada, 2020.

M. Lindfors, T. Chen, and C. A. Naesseth. Robust Gaussian process regression with G-confluent likelihood. In *21th IFAC World Congress*, Germany, 2020.

C. A. Naesseth, F. Lindsten, and T. B. Schön. Elements of sequential Monte Carlo. *Foundations and Trends® in Machine Learning*, 12(3):307–392, November 2019a. Now Publishers, Inc.

C. A. Naesseth, F. Lindsten, and T. B. Schön. High-dimensional filtering using nested sequential Monte Carlo. *IEEE Transactions on Signal Processing*, 67(16):4177–4188, August 2019b.

C. A. Naesseth. *Machine learning using approximate inference: Variational and sequential Monte Carlo methods*. PhD thesis, Linköping University, 2018. **(Savage Award for outstanding dissertation in Theory and Methods)**.

D. Lawson, G. Tucker, C. A. Naesseth, C. J. Maddison, R. P. Adams, and Y. W. Teh. Twisted variational sequential Monte Carlo. In *Third workshop on Bayesian Deep Learning (NeurIPS)*, Montreal, Canada, 2018.

C. A. Naesseth, S. W. Linderman, R. Ranganath, and D. M. Blei. Variational sequential Monte Carlo. In *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics (AISTATS)*, Lanzarote, Spain, Apr 2018.

C. A. Naesseth, F. J. R. Ruiz, S. W. Linderman, and D. M. Blei. Reparameterization gradients through acceptance–rejection algorithms. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, Fort Lauderdale, USA, Apr 2017. **(Best Paper Award)**.

F. Lindsten, A. M. Johansen, C. A. Naesseth, B. Kirkpatrick, T. B. Schön, J. Aston, and A. Bouchard-Côté. Divide-and-conquer with sequential Monte Carlo. *Journal of Computational and Graphical Statistics*, 2016.

T. Rainforth*, C. A. Naesseth*, F. Lindsten, B. Paige, J-W. van de Meent, A. Doucet, and F. Wood. Interacting particle Markov chain Monte Carlo. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, New York, USA, Jun 2016. * equal contribution.

C. A. Naesseth, F. Lindsten, and T. B. Schön. Towards automated sequential Monte Carlo methods for probabilistic graphical models. In *NIPS Workshop on Black Box Learning and Inference*, Montreal, Canada, 2015a.

T. B. Schön, F. Lindsten, J. Dahlin, J. Wågberg, C. A. Naesseth, A. Svensson, and L. Dai. Sequential Monte Carlo Methods for System Identification. In *Proceedings of the 17th IFAC Symposium on System Identification (SYSID)*, Beijing, China, 2015.

C. A. Naesseth, F. Lindsten, and T. B. Schön. Nested Sequential Monte Carlo Methods. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, Lille, France, Jul 2015b.

C. A. Naesseth, F. Lindsten, and T. B Schön. Sequential Monte Carlo for Graphical Models. In *Advances in Neural Information Processing Systems (NIPS) 27*, pages 1862–1870, Montreal, Canada, 2014a.

C. A. Naesseth, F. Lindsten, and T. B. Schön. Capacity estimation of two-dimensional channels using sequential Monte Carlo. In *Proceedings of the 2014 IEEE Information Theory Workshop (ITW)*, pages 431–435, Hobart, Australia, Nov 2014b.